Intro to Accessibility and Aging

IN4MATX 232

Yiqi Xiao with the help of Gemini

What Is **Accessibility**?

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Accessibility is broadly defined as the degree to which ***a product, device, service, or environment is available to as many people as possible***. In the context of digital systems, the World Intellectual Property Organization (WIPO) and the International Standards Organization (ISO) define accessibility in terms of the prevention of activity limitations and the facilitation of participation for people with disabilities. Historically, accessibility was viewed as a ***technical compliance issue rooted in the medical model of disability***.

What Is **Assistive Technology**?

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Assistive Technology has traditionally been defined through a legislative and medical lens. The Technology-Related Assistance for Individuals with Disabilities Act of 1988 (the "Tech Act") provided the first official definition in the United States, describing AT as ***any item, piece of equipment, or product system used to increase, maintain, or improve the functional capabilities of individuals with disabilities***.

The reauthorization of the Assistive Technology Act in 2004 marked a key shift, ***placing individuals with disabilities at the forefront of service planning and expanding the scope to include "electronic and information technology".*** Today, the WHO utilizes AT as an "umbrella term" encompassing both products and the related systems and services that help maintain functioning in cognition, communication, hearing, mobility, self-care, and vision.

(Special) Education

Geriatric Medicine

Disability Studies

Biomedical Engineering

Inclusive Design

Audiology & Speech-Language Pathology

Accessibility & Aging

Human-Computer Interaction

Neuropsychology

Urban Planning & Architecture

Clinical Psychology

Public Policy

Assistive Technology

The History of the Accessibility and Aging Subcommittee at CHI

Year	Subcommittee Name	Relationship to Aging/Accessibility
2009-2013	Usability, Accessibility and User Experience	Integrated with Usability; Aging in "Specific Applications".
2014-2016	Specific Application Areas	Accessibility and Aging both moved to "Specific Applications".
2017-2018	Health, Accessibility and Aging	Combined Health and Accessibility into a dedicated track.
2019-Present	Accessibility and Aging	Standalone subcommittee for Accessibility and Aging.

CHI 2026

This subcommittee is suitable for contributions related to the design or study of technology for people with disabilities and/or older adults. Accessibility papers…technology designed for or used by people with disabilities ***including sensory, motor, mobility, psychosocial or cognitive, intellectual or learning disabilities, or people who identify as neurodivergent.*** Aging papers are …used by ***people in the later stages of life***….We welcome contributions across a range of topics aimed at benefiting relevant stakeholder groups and not solely limited to concerns of making technology accessible.

CHI 2026 (continued)

Note that if your paper primarily concerns health outcomes or interactions with health data or with healthcare providers, then the Health subcommittee is probably a better fit, whereas papers reflecting on how technologies are used or designed for specific needs are a better fit for this subcommittee….***This subcommittee welcomes all contributions related to accessibility and aging, including empirical, theoretical, conceptual, methodological, design, and systems contributions.***

In-Scope vs Out-of-Scope

Health Outcomes: At CHI, if a paper primarily concerns health outcomes or interactions with health data/providers, it is generally considered a better fit for the Health subcommittee.

Beyond Technology: The CHI subcommittee welcomes papers that are "not solely limited to concerns of making technology accessible," such as studies on how older adults learn programming or choose not to participate in social interventions.

CHI 2026 (continued)

Your paper should also use **inclusive language**. Avoid characterizing an entire population using phrases that represent outlier positions for individuals, like “suffering from” (negative) or “inspirational” (positive). When comparing across characteristics or experiences, avoid referring to some people as “normal” implying others are not, or broadly characterizing one group’s experiences as categorically better or worse than others. We also recommend avoiding terms like “vulnerable,” “special needs,” and “X challenged.” We understand there may be exceptions (e.g., medical vision classification systems define a visual acuity as “normal”), but such usage should be footnoted for clarity, and these measures should be relevant to the described work.

Examples:

- Discovering Accessible Data Visualisations for People with ADHD
- Analyzing Accessibility Reviews Associated with Visual Disabilities or Eye Conditions
- Envisioning the (In)Visability of Discreet and Wearable AAC Devices
- Technology Adoption and Learning Preferences for Older Adults: Evolving Perceptions, Ongoing Challenges, and Emerging Design Opportunities
- Screen Recognition: Creating Accessibility Metadata for Mobile Applications from Pixels
- Understanding Older Adults' Participation in Design Workshops
- The Promise of Empathy: Design, Disability, and Knowing the "Other"
- "If It's Important It Will Be A Headline": Cybersecurity Information Seeking in Older Adults
- Older People Inventing their Personal Internet of Things with the IoT Un-Kit Experience
- Addressing Age-Related Bias in Sentiment Analysis

ASSETS

The International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) remains the undisputed premier forum for ***research concerning the design, evaluation, use, and education related to computing for people with disabilities and older adults***. Since its inception in 1994, ASSETS has fostered an environment where high-quality original submissions can explore both mainstream and specialized technologies.

ASSETS

Submission Track	Primary Objective
Technical Papers (Long)	Present mature, significant contributions to design, systems, or scientific understanding.
Technical Papers (Short)	Focused research contributions, such as a novel interaction technique or a pilot study.
Experience Reports	Reflection on the application of accessibility in education, industry, or policy settings.
Posters and Demos	Interactive showcasing of prototypes or preliminary research findings.
Doctoral Consortium	Mentorship for PhD students to refine their research focus and methodology.
Student Research Comp.	Competition for students to present original research in a professional setting.

CSCW

The ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) approaches accessibility through the lens of social interaction and collaboration. For a paper to be in scope for CSCW, it must focus on "social aspects of technology mediation" and be properly contextualized in the CSCW literature, ***referencing concepts such as collective intelligence, crowdsourcing, or remote collaboration***. A critical distinction at CSCW is the definition of "out of scope" work. Research contributions that are ***primarily of relevance to individual users are considered out of scope***. This stands in contrast to ASSETS, which frequently accepts work focusing on individual assistive technologies.

Compare CHI, CSCW, ASSETS?

Conference Other Fields?

AAAI Conference on Artificial Intelligence

AAAI/ACM Conference on AI, Ethics, and Society (AIES)

American Medical Informatics Association (AMIA) Annual Symposium

CHI PLAY

ISLS

…

Key Trends in this Field

From CS & Medicine → Human-centered fields focused on equity & justice

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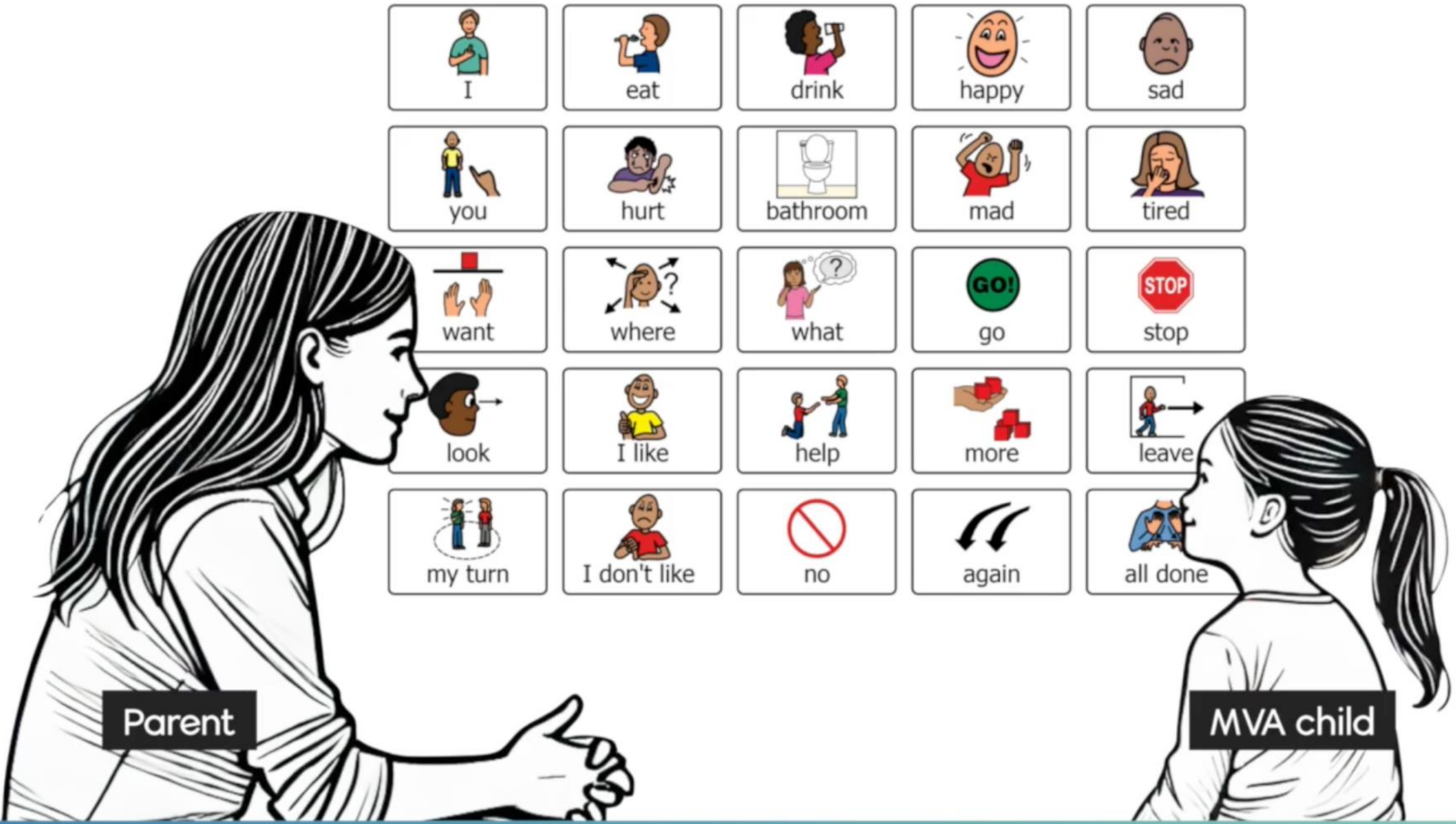
Toward participatory research: "Nothing about us without us"

Challenges in this Field

Background

To better understand their child's thoughts, many families adopt **Augmentative and Alternative Communication (AAC)** tool.




Hello!
ABC 123
PEOPLE
QUESTION
ACTIONS
SOCIAL
PLACES
TIME
GROUPS
DESCRIBE
good
clear
I
me
to
come
that
a-
the
and
more
.
my
is
eat
drink
Finish
get
all
at
COLORS
EXTRA WORDS
it
can
go
help
open
put
in
for
on
here
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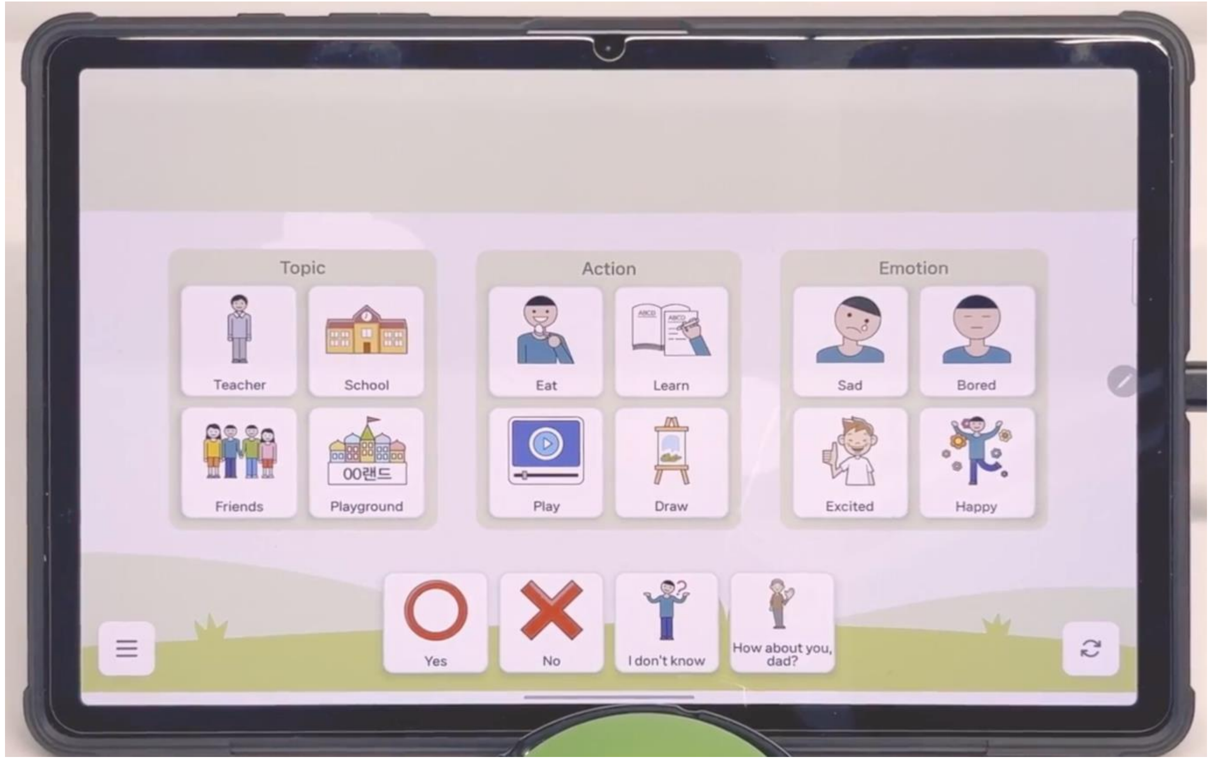
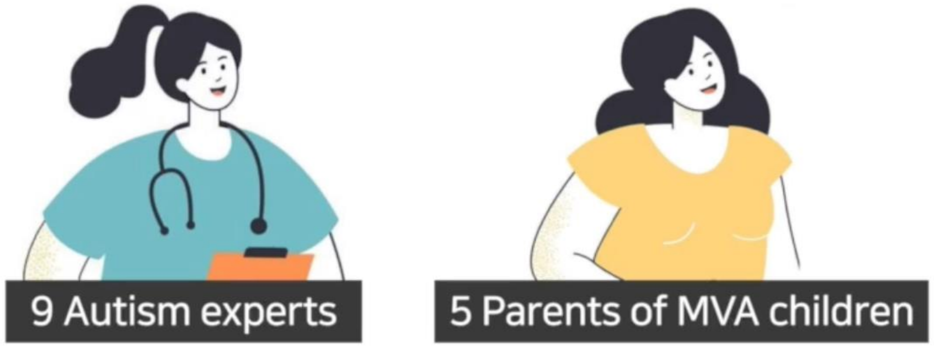

Topic
Action
Emotion
Teacher
School
Eat
Learn
Sad
Bored
Friends
00랜드
Playground
Play
Draw
Excited
Happy
Yes
No
I don't know
How about you, dad?

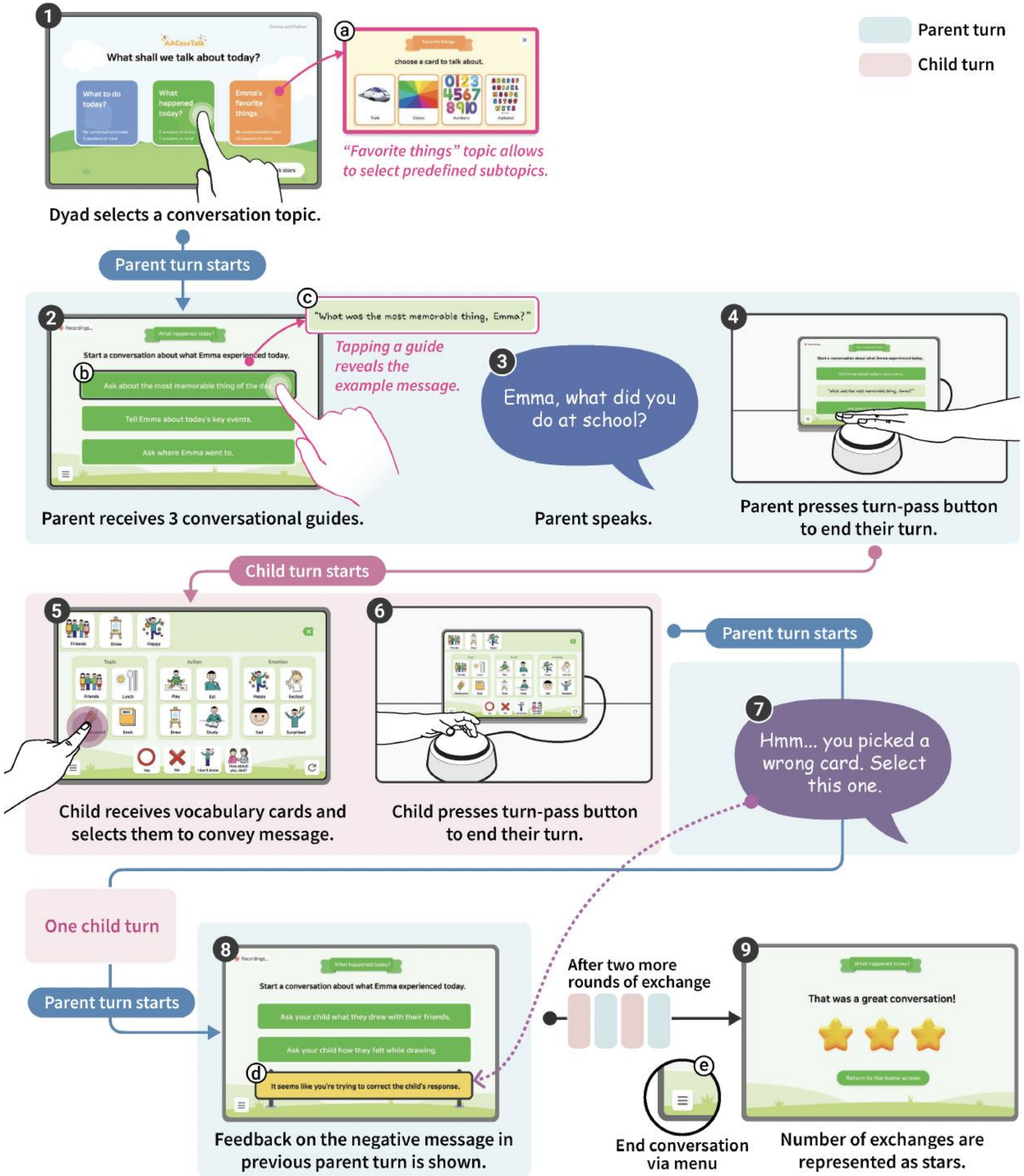


Design Rationales

DR1. Structure Turn-taking through Expressive Cues.

DR2. Contextualize Parental Guidance.

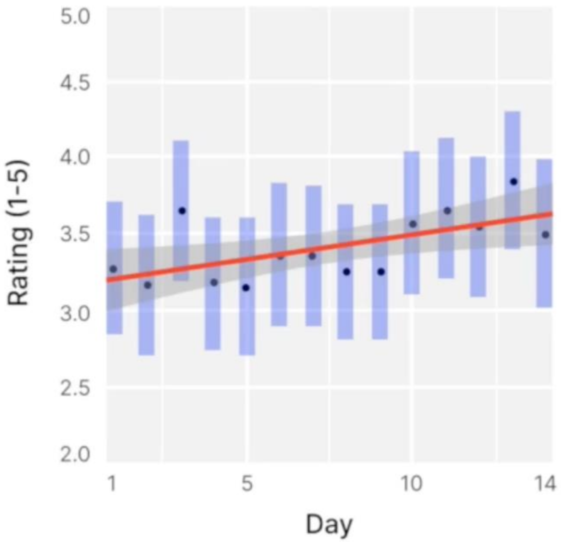
DR3. Prioritize MVA Children's Vocabulary.


Parent turn
Child turn
What shall we talk about today?
What to do today?
What happened today?
Emma's favorite things
choose a card to talk about.
"Favorite things" topic allows to select predefined subtopics.
Dyad selects a conversation topic.
Parent turn starts
"What was the most memorable thing, Emma?"
Start a conversation about what Emma experienced today.
Ask about the most memorable thing of the day.
Tell Emma about today's key events.
Ask where Emma went to.
Tapping a guide reveals the example message.
Emma, what did you do at school?
Parent receives 3 conversational guides.
Parent speaks.
Parent presses turn-pass button to end their turn.
Child turn starts
Parent turn starts
Hmm... you picked a wrong card. Select this one.
Child receives vocabulary cards and selects them to convey message.
Child presses turn-pass button to end their turn.
One child turn
Parent turn starts
Start a conversation about what Emma experienced today.
Ask your child what they drew with their friends.
Ask your child how they felt while drawing.
It seems like you're trying to correct the child's response.
After two more rounds of exchange
That was a great conversation!
Return to the home screen
End conversation via menu
Feedback on the negative message in previous parent turn is shown.
Number of exchanges are represented as stars.

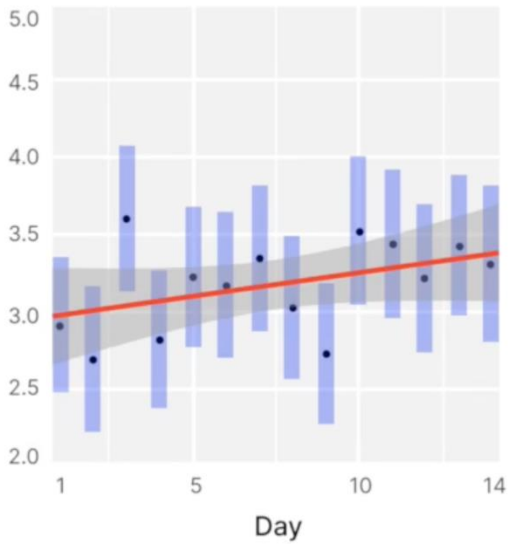
Findings

Overall System Usage

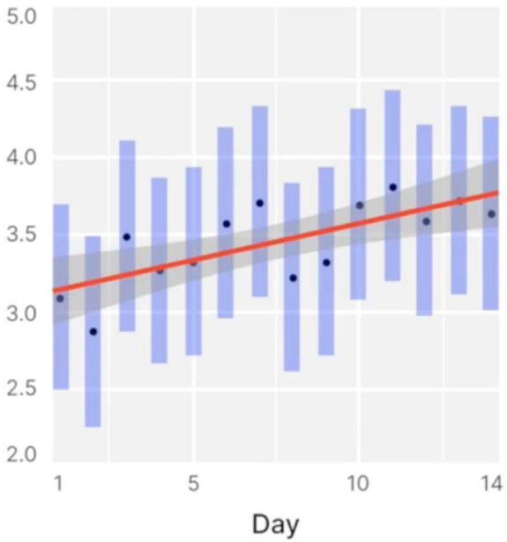
Improved Interaction quality



(a) Overall satisfaction with the conversation



(b) Smoothness of turn-taking

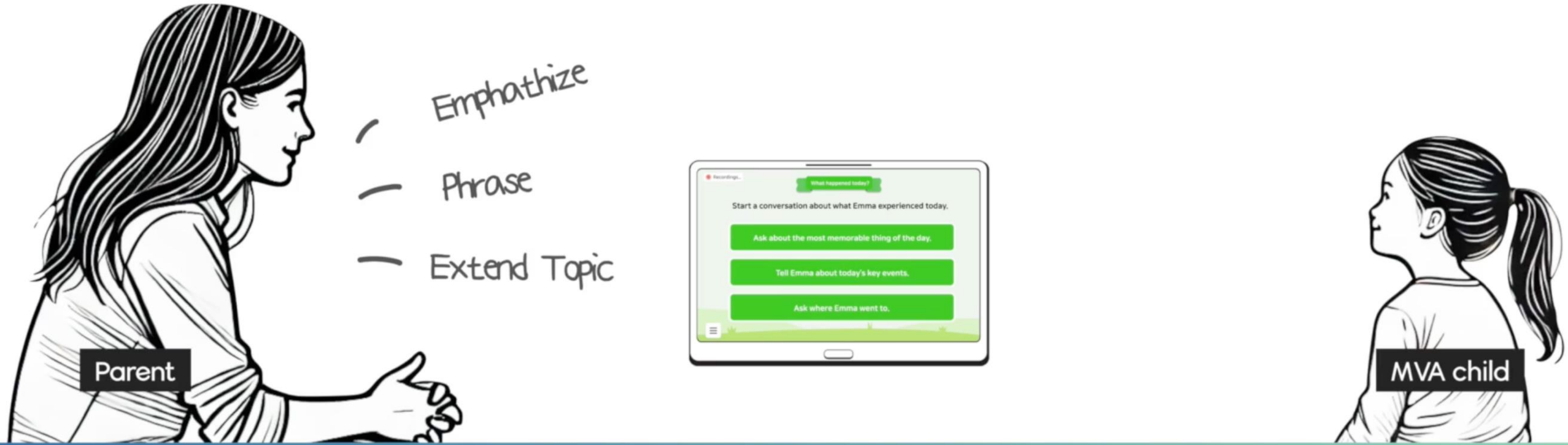


(c) Level of child engagement

Findings

AACessTalk helped parents break out of repetitive conversations patterns.

With conversational guides, parents explored new ways to engage, making conversations more dynamic.



Findings

AACessTalk helped parents break out of repetitive conversations patterns.

Parents can recognize their child's broader language abilities and engage in richer conversations.

	[...]
P6	The drone flew up into the sky. How did you feel while flying it?
C6	Balloon Sky Button Sky Scared
P6	Scared? Were you scared flying the drone, [child]?

Dialogue 1. Dialogue snippet from P6 and C6.

Findings

AACessTalk gave children more control over communication.

Parents became more aware of their child's communicative intent.

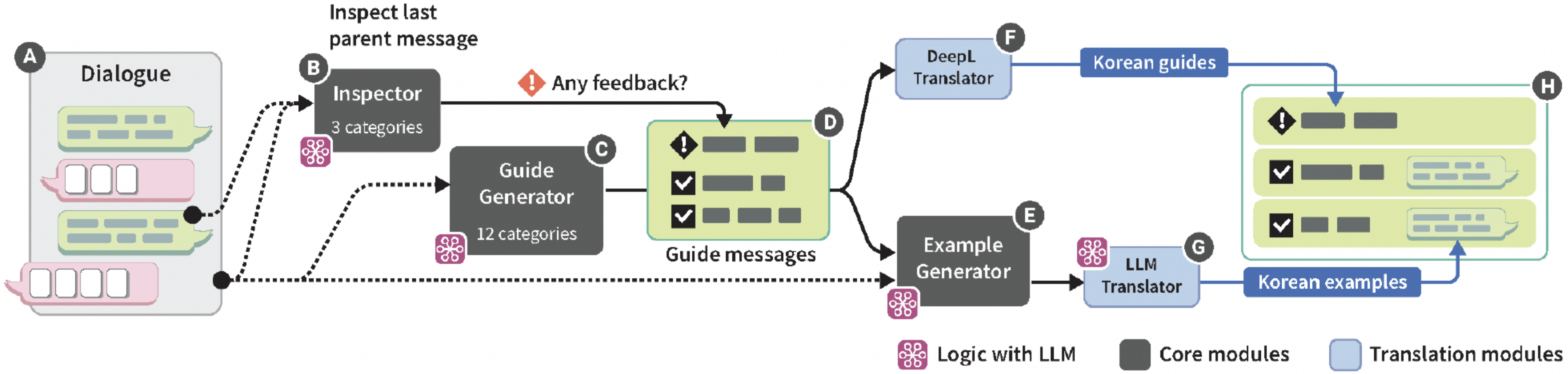
Initiated conversation
by handing the tablet

Ended conversations
by closing the tablet

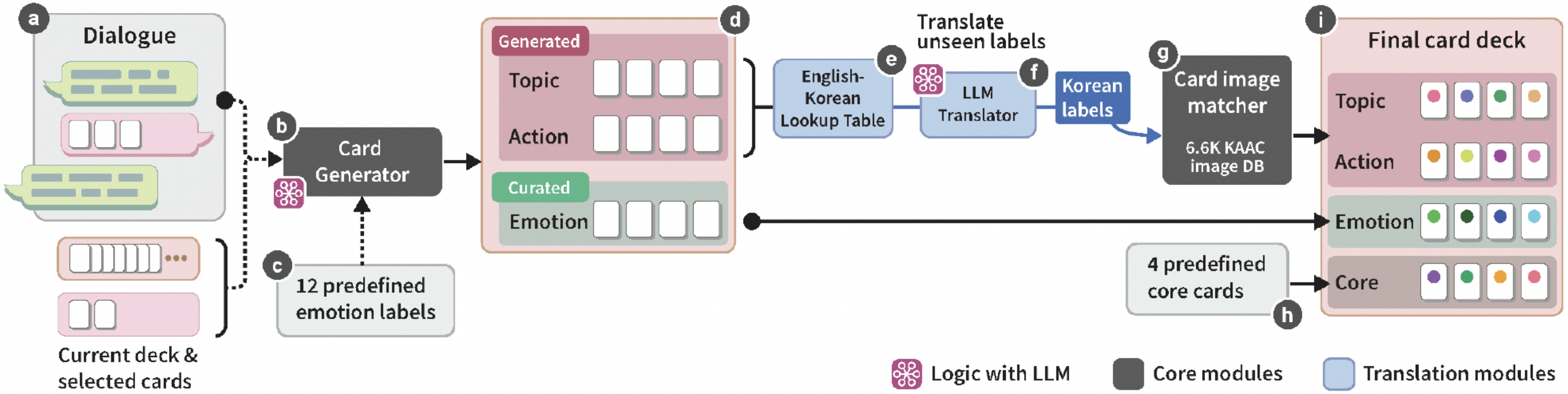
Asserting their turn

What does this this project make you think?

AI's role?



Dialogue
Inspect last parent message
Inspector
3 categories
Any feedback?
Guide Generator
12 categories
Guide messages
DeepL Translator
Korean guides
Example Generator
LLM Translator
Korean examples
Logic with LLM
Core modules
Translation modules



Dialogue
Current deck & selected cards
Card Generator
12 predefined emotion labels
Generated
Topic
Action
Curated
Emotion
English-Korean Lookup Table
Translate unseen labels
LLM Translator
Korean labels
Card image matcher
6.6K KAAC image DB
4 predefined core cards
Final card deck
Topic
Action
Emotion
Core
Logic with LLM
Core modules
Translation modules

Is this strength-based?

Ability-based Approach

*"Ability-based design consists of **focusing on ability throughout the design process** in an effort to create systems that leverage the full range of human potential."*

" In making the shift to ability-based design, we move away from assisting human users to conform to inflexible computer systems, and instead consider how systems can be made to fit the abilities of whoever uses them.."

Wobbrock, J. O., Kane, S. K., Gajos, K. Z., Harada, S., & Froehlich, J. (2011). *Ability-Based Design: Concept, Principles and Examples*.

Strength-based Approach

"The strengths perspective is a dramatic departure from the conventional way of doing social work. It is a versatile and effective approach that **focuses on the strengths and resources of people and their environments, rather than on their problems and pathologies**."

“All people possess a wide range of **talents, abilities, capacities, skills, resources, and aspirations**… a belief in human potential is tied to the notion that people have **untapped, undetermined reservoirs** of mental, physical, emotional, social, and spiritual abilities.”

Saleebey, D. (1992). *The Strengths Perspective in Social Work Practice*.

Interdependent

*"Communication inherently involves **interdependent** actions that require mutual effort from all parties. Bennett et al. [13] applied the concept of interdependence from Disability Studies to the design of assistive technology (AT), proposing that AT needs to support collaborative efforts among people with disabilities, those they interact with, and their surrounding environments. Through the lens of interdependence, current communication technologies for MVA children-parent dyads predominantly focus on facilitating access for MVA children."*

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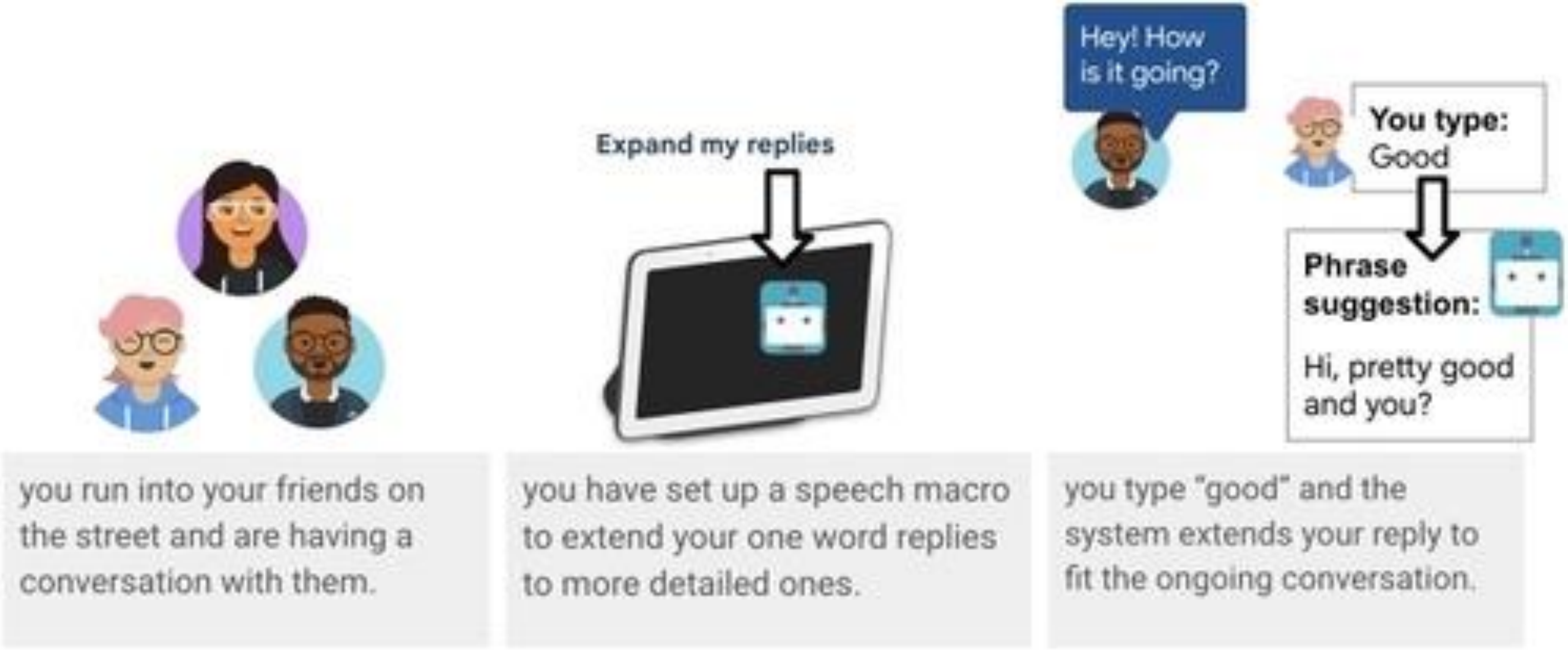
From independent → interdependent

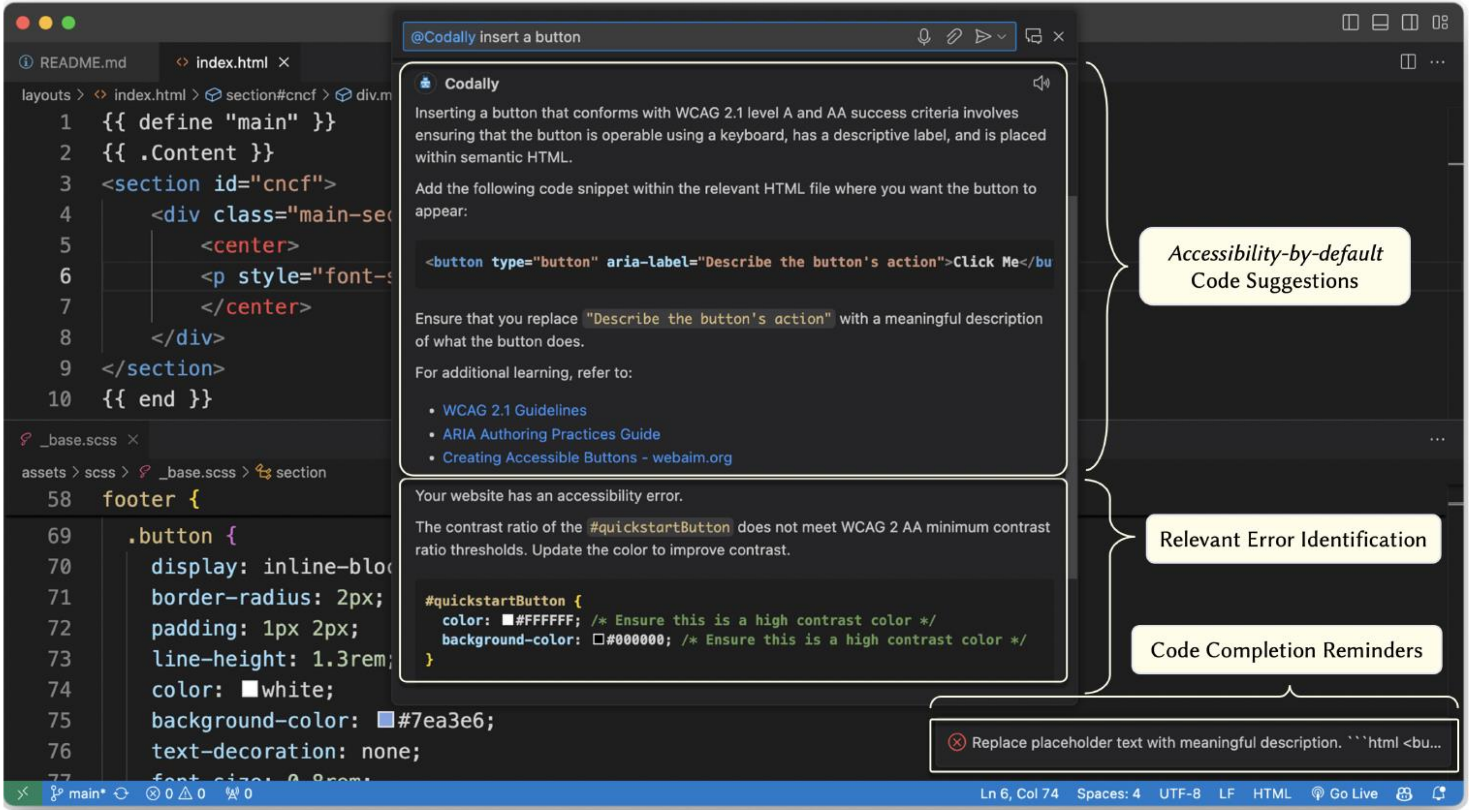
More work about AAC with AI



Valencia, S., Kallarackal, K., Mankoff, J., et al. (2023).
"The less I type, the better": How AI language models can enhance or impede communication for AAC users.
Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, Article 830, 1–14.
https://doi.org/10.1145/3544548.3581560

Weinberg, T. M., Valencia, S., Kadoma, K., et al. (2025).
Why so serious? Exploring timely humorous comments in AAC conversations.
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, Article 759, 1–19.
https://doi.org/10.1145/3706598.3714102

CodeA11y: Making AI Coding Assistants Useful for Accessible Web Development



Let’s compare them!

What literature does each paper draw from?

Problem framing

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AACessTalk

“Minimally verbal autistic (MVA)1 children engage in communication mainly with nonverbal cues… parents often shoulder the significant responsibility of driving this communication… they experience feelings of isolation and frustration.”

↓

3 design rationales for AIs as communication mediators resulted from formative study with parents and experts.

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CodeA11y

"Most websites contain extensive accessibility errors, despite decades of investment in standards and guidelines, tools, advocacy, and policy."

↓

"Our formative study... revealed three key issues in AI-assisted coding: failure to prompt AI for accessibility, omitting crucial manual steps like replacing placeholder attributes, and the inability to verify compliance."

Who are they (each intervention) designed for?

Which disability models do they each align with?

Let’s talk about conferences a bit more (extend from last class!

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And identifying other resources?